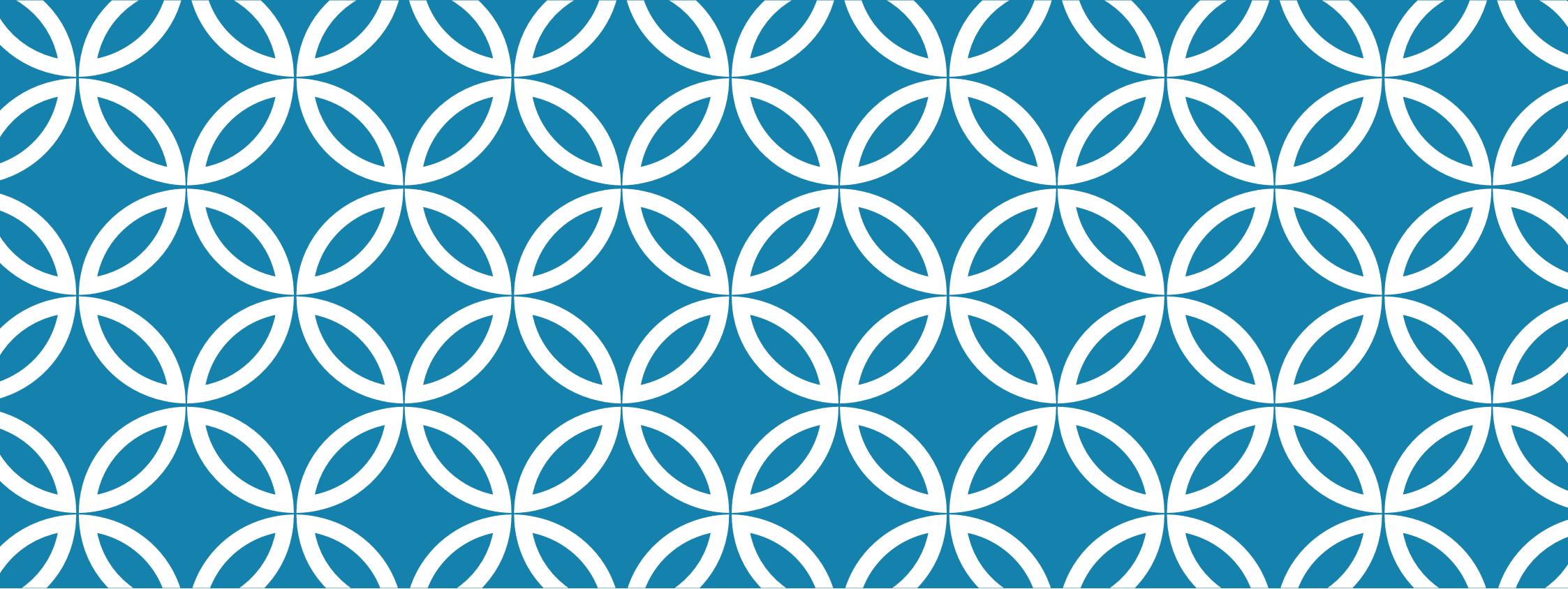




DEVELOPMENT OF URINARY SYSTEM



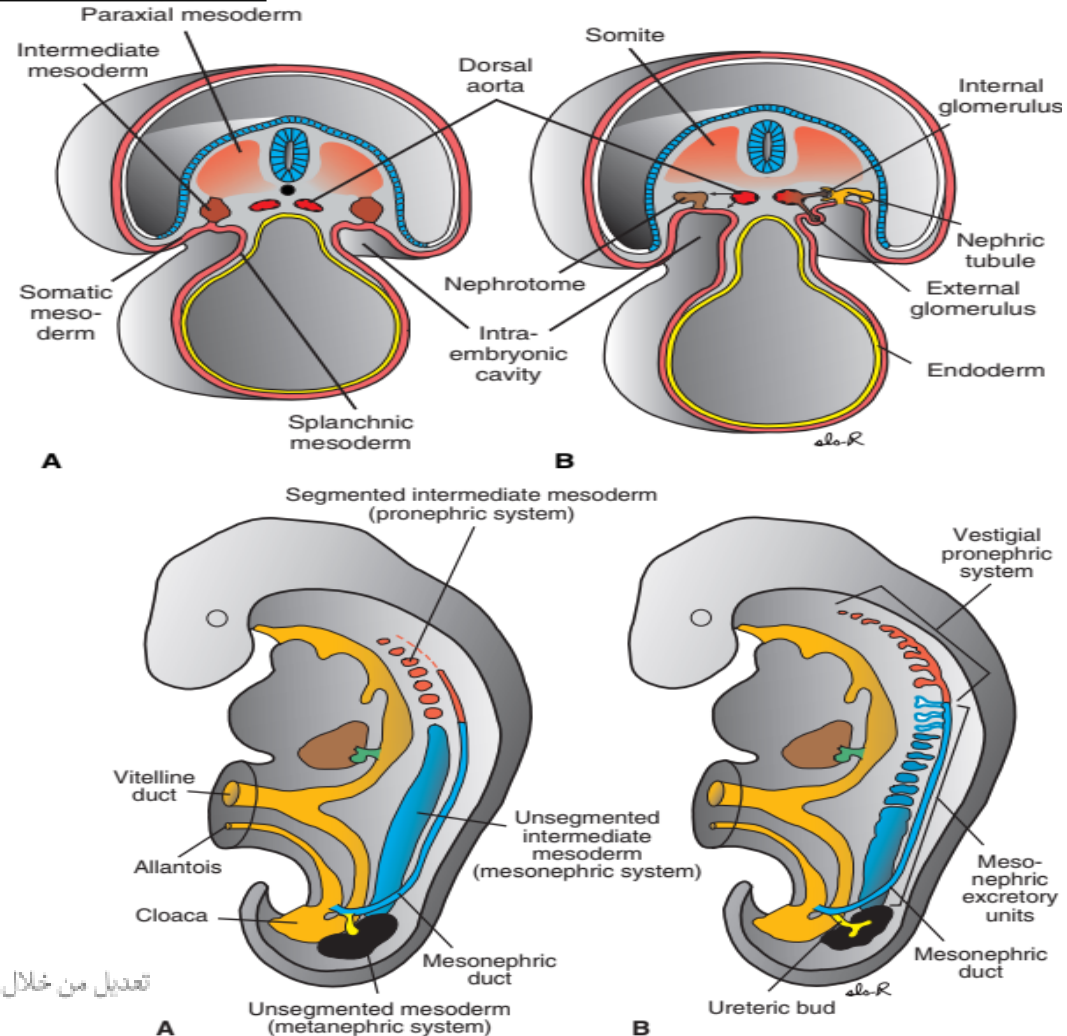
تعديل من خلال WPS Office

M. G. HAMMAD

URINARY SYSTEM

KIDNEY SYSTEM

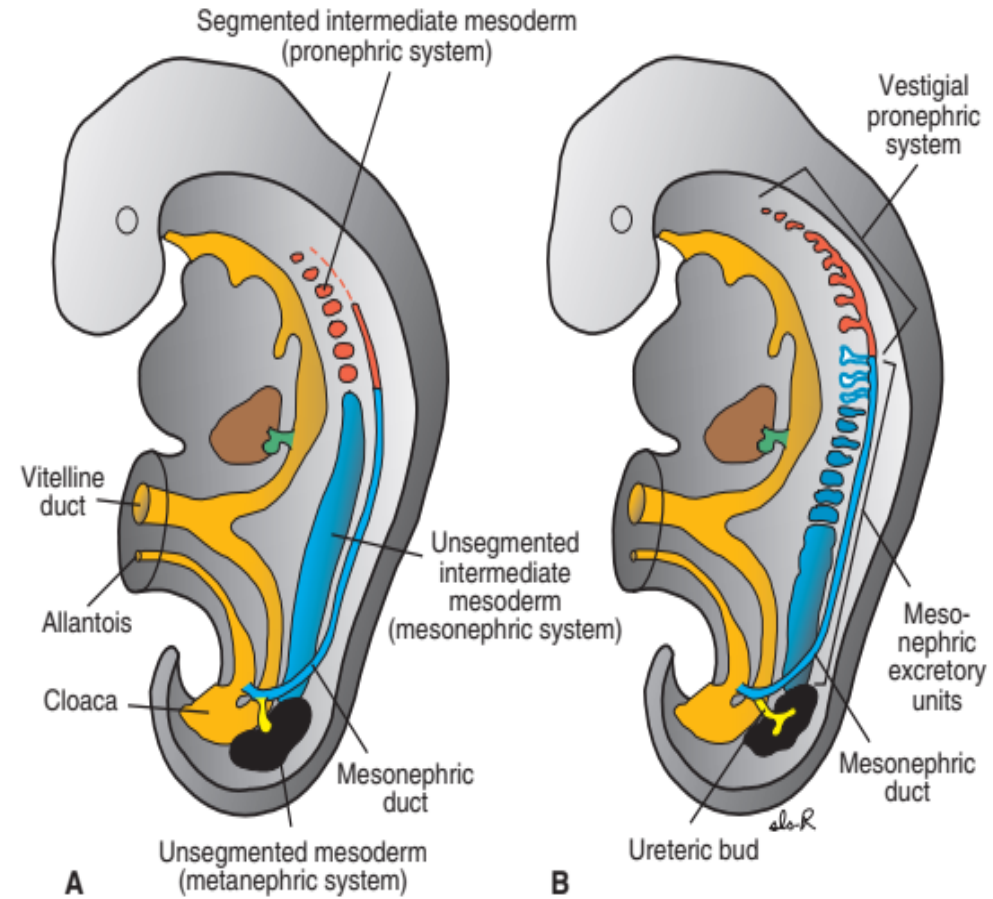
- ❑ Renal system developed from intermediate mesoderm.
- ❑ Along the posterior wall of the abdominal cavity.
- ❑ Developmentally, kidney system composed of 3 parts:
 - i. **Pronephros:** rudimentary and nonfunctional.
 - ii. **Mesonephros:** may function for a short time during the early fetal period.
 - iii. **Metanephros:** forms the permanent kidney.



KIDNEY SYSTEM

PRONEPHROS

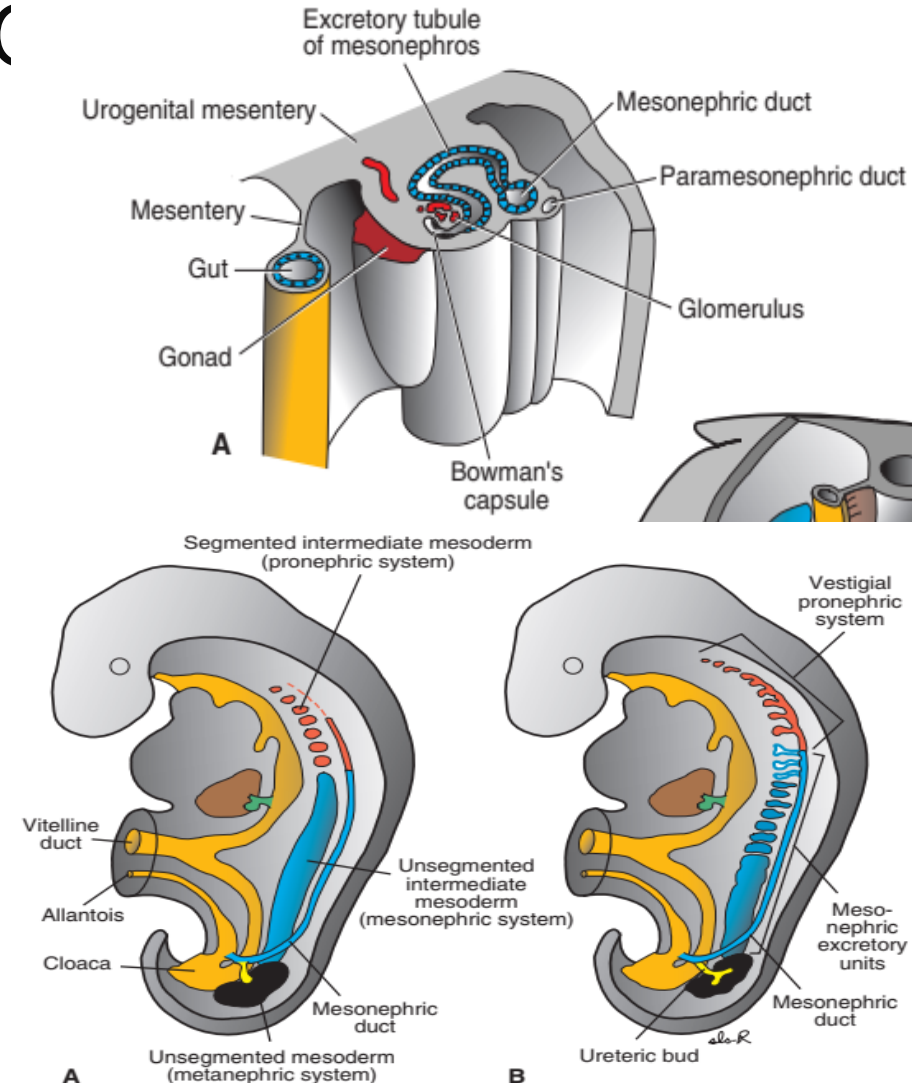
- ❑ Appears at beginning of 4th week.
- ❑ Presents as 7 to 10 solid cell groups in the cervical region.
- ❑ These groups form excretory units (nephrotomes).
- ❑ They regress before more caudal ones are formed.
- ❑ By the end of the fourth week, all pronephric system have disappeared.



KIDNEY SYSTEM

MESONEPHROS

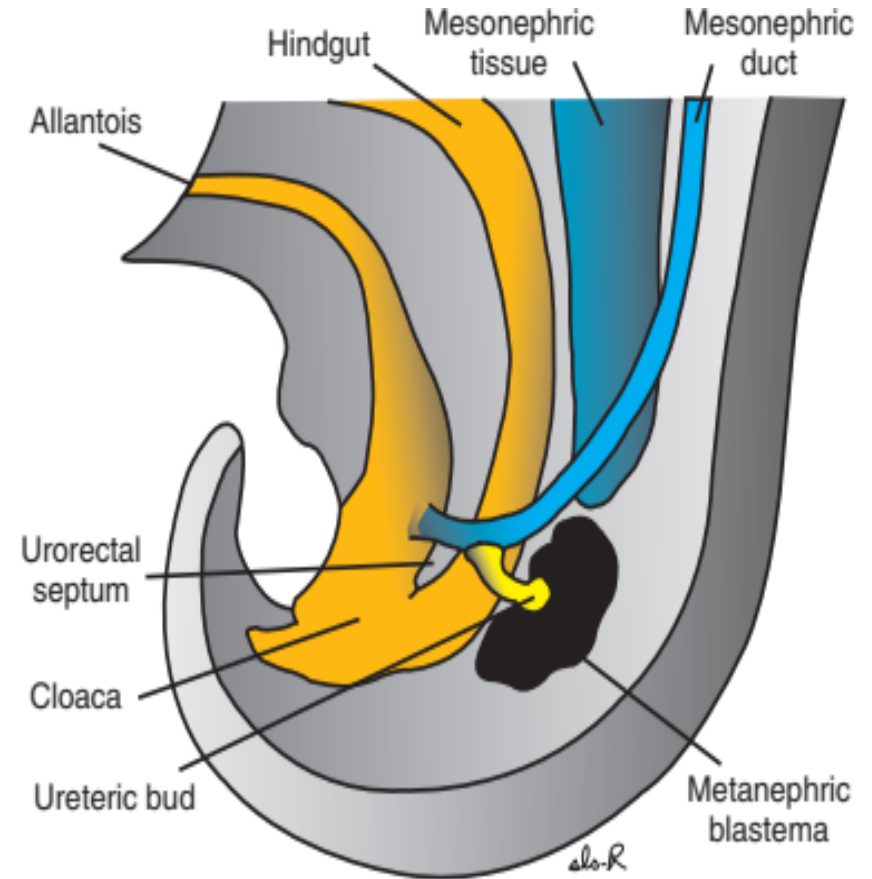
- ❑ Mesonephros and mesonephric ducts are derived from intermediate mesoderm.
- ❑ From upper thoracic to upper lumbar (L3) segments.
- ❑ Appears early in 4. week.
- ❑ It lengthen rapidly.
- ❑ Form an S-shaped loop and acquire a tuft of capillaries.
- ❑ Capillaries form a glomerulus at their medial extremity.
- ❑ Around the glomerulus, the tubules form Bowman's capsule.
- ❑ Now these 2 structures called renal corpuscle.
- ❑ Laterally, the tubule enters the longitudinal collecting duct known as the mesonephric or wolffian duct.



KIDNEY SYSTEM

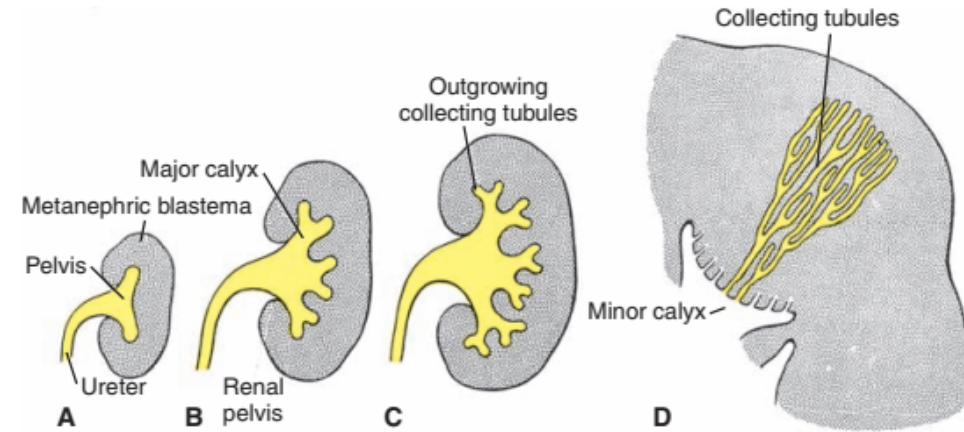
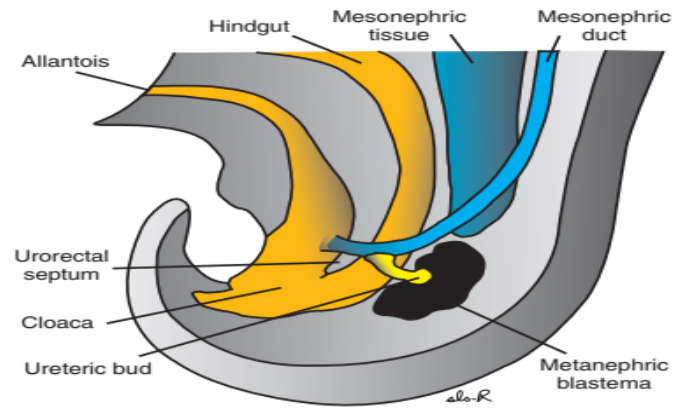
METANEPHROS (DEFINITIVE KIDNEY)

- ❑ Appears in the fifth week.
- ❑ Its excretory units develop from metanephric mesoderm.
- ❑ The development of the duct system differs from that of the other kidney system.



DEFINITIVE KIDNEY COLLECTING DUCTS SYS

- ❑ Develop from the ureteric bud.
- ❑ Ureteric bud: an outgrowth of the mesonephric duct close to its entrance to the cloaca.
- ❑ Subsequently, the bud dilates, forming (primitive renal pelvis).
- ❑ Then splits into cranial and caudal portions (the future major calyces).
- ❑ Each calyx forms two new buds while penetrating the metanephric tissue.
- ❑ These buds continue to subdivide until 12 or more generations of tubules have formed.
- ❑ Now minor calyces & renal pyramids are formed.

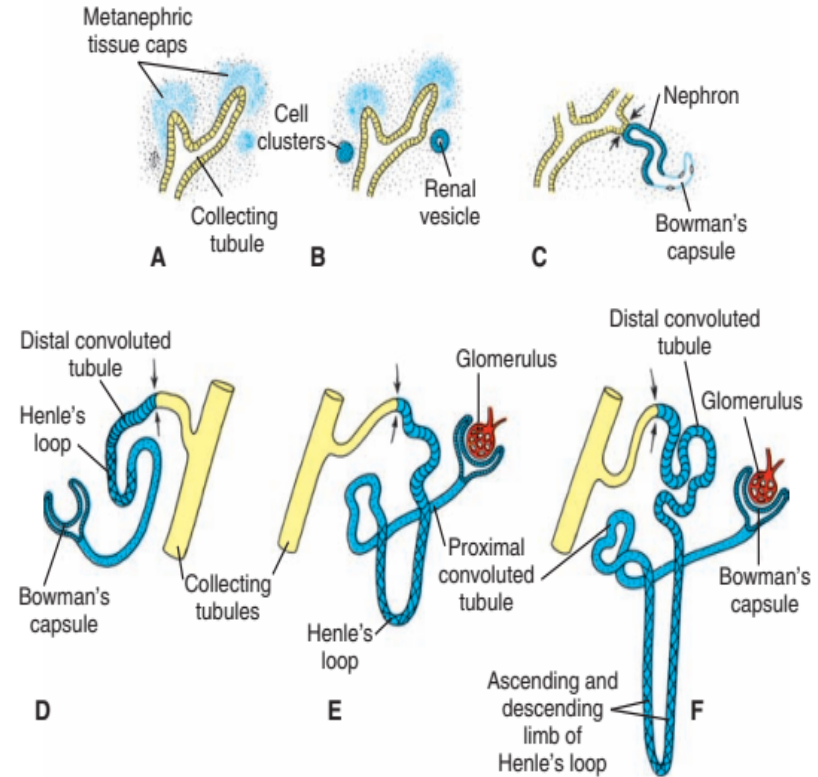


❑ N.B:

- ✓ Ureteric bud gives rise to:
 - Ureter.
 - Renal pelvis.
 - Major and minor calyces.
 - Approximately 1 to 3 million collecting tubules.

DEFINITIVE KIDNEY EXCRETORY DUCTS SYSTEM

- ❑ Each newly formed collecting tubule is covered at its distal end by a metanephric tissue cap.
- ❑ Cells of the tissue cap form small vesicles (renal vesicles).
- ❑ Renal vesicles give rise to small S-shaped tubules.
- ❑ Capillaries grow into the pocket at one end of the S and differentiate into glomeruli.
- ❑ Tubules together with their glomeruli, form nephrons, or excretory units.
- ❑ The proximal end of each nephron forms Bowman's capsule.
- ❑ The distal end forms an open connection with one of the collecting tubules.
- ❑ Continuous lengthening of the excretory tubule results in formation of:
 - ✓ Proximal convoluted tubule.
 - ✓ Loop of Henle.
 - ✓ Distal convoluted tubule.



Renal Tumors and Defects

Wilms' tumor is a cancer of the kidneys that usually affects children by 5 years of age but may also occur in the fetus. Wilms' tumor is due to mutations in the *WT1* gene on 11p13, and it may be associated with other abnormalities and syndromes. For example, **WAGR syndrome** is characterized by Wilms' tumor, aniridia, gonadoblastomas (tumors of the gonads), and mental retardation (intellectual disability). The constellation of defects is due to a microdeletion in chromosome 11 that includes both the *PAX6* (aniridia) and *WT1* genes that are only 700 kb apart. Similarly, **Denys-Drash syndrome** consists of renal failure, ambiguous genitalia, and Wilms' tumor.

Renal dysplasias and **agenesis** are a spectrum of severe malformations that represent the primary diseases requiring dialysis and transplantation in the first years of life. **Multicystic dysplastic kidney** is one example of this group of abnormalities in which numerous ducts are surrounded by undifferentiated cells. Nephrons fail to develop, and the ureteric bud fails to branch, so that the collecting ducts never form. In some cases, these defects cause involution of the kidneys and **renal agenesis**. Renal agenesis may also arise if the interaction between the metanephric mesoderm and the ureteric bud fails to occur. Normally, during the interaction, **GDNF** produced by the metanephric mesoderm produces branching and growth of the ureteric bud. Thus, mutations in genes that regulate *GDNF* expression of signaling may result in renal agenesis. Examples include the gene *SALL1*, responsible for Townes-Brock syndrome; *PAX2* that causes renal coloboma syndrome; and *EYA1* that results in branchio-oto-renal syndrome. Bilateral renal agenesis, which occurs in 1/10,000 births, results in renal failure. The baby presents with **Potter sequence**, characterized by anuria, oligohydramnios (decreased volume of amniotic fluid), and hypoplastic lungs secondary to the oligohydramnios. In 85% of cases, other severe defects, including absence or abnormalities of the vagina and uterus, vas deferens, and seminal vesicles, accompany this condition. Common associated defects in other systems include cardiac anomalies, tracheal and duodenal atresias, cleft lip and palate, and brain abnormalities. Because of the oligohydramnios, the uterine cavity is compressed resulting in a characteristic appearance of the fetus, including a flattened face (Potter facies) and club feet.

In **congenital polycystic kidney disease** (Fig. 16.8), numerous cysts form. It may be inherited as an autosomal recessive or autosomal

dominant disorder or may be caused by other factors. **Autosomal recessive polycystic kidney disease (ARPKD)**, which occurs in 1/5,000 births, is a progressive disorder in which cysts form from collecting ducts. The kidneys become very large, and renal failure occurs in infancy or childhood. In **autosomal dominant polycystic kidney disease (ADPKD)**, cysts form from all segments of the nephron and usually do not cause renal failure until adulthood. The autosomal dominant disease is more common (1/500 to 1/1,000 births), but less progressive than the autosomal recessive disease. Both types of disease are linked to mutations in genes that encode proteins localized in cilia and that are important for ciliary function. These abnormalities belong to a growing group of diseases called the **ciliopathies** that are due to mutations in cilia related proteins. These disorders include **Bardet-Biedal syndrome**, characterized by renal cysts, obesity, intellectual disability, and limb defects, and **Meckel Gruber syndrome**, characterized by renal cysts, hydrocephalus, microphthalmia, cleft palate, absence of the olfactory tract, and polydactyly. Since cilia are present on most cell types and in most tissues, many organ systems can be affected by abnormalities in ciliary structure and function.

Duplication of the ureter results from early splitting of the ureteric bud (Fig. 16.9). Splitting may be partial or complete, and metanephric tissue may be divided into two parts, each with its own renal pelvis and ureter. More frequently, however, the two parts have a number of lobes in common as a result of intermingling of collecting tubules. In rare cases, one ureter opens into the bladder, and the other is ectopic, entering the vagina, urethra, or vestibule (Fig. 16.9C). This abnormality results from development of two ureteric buds. One of the buds usually has a normal position, whereas the abnormal bud moves down together with the mesonephric duct. Thus, it has a low, abnormal entrance in the bladder, urethra, vagina, or epididymal region.

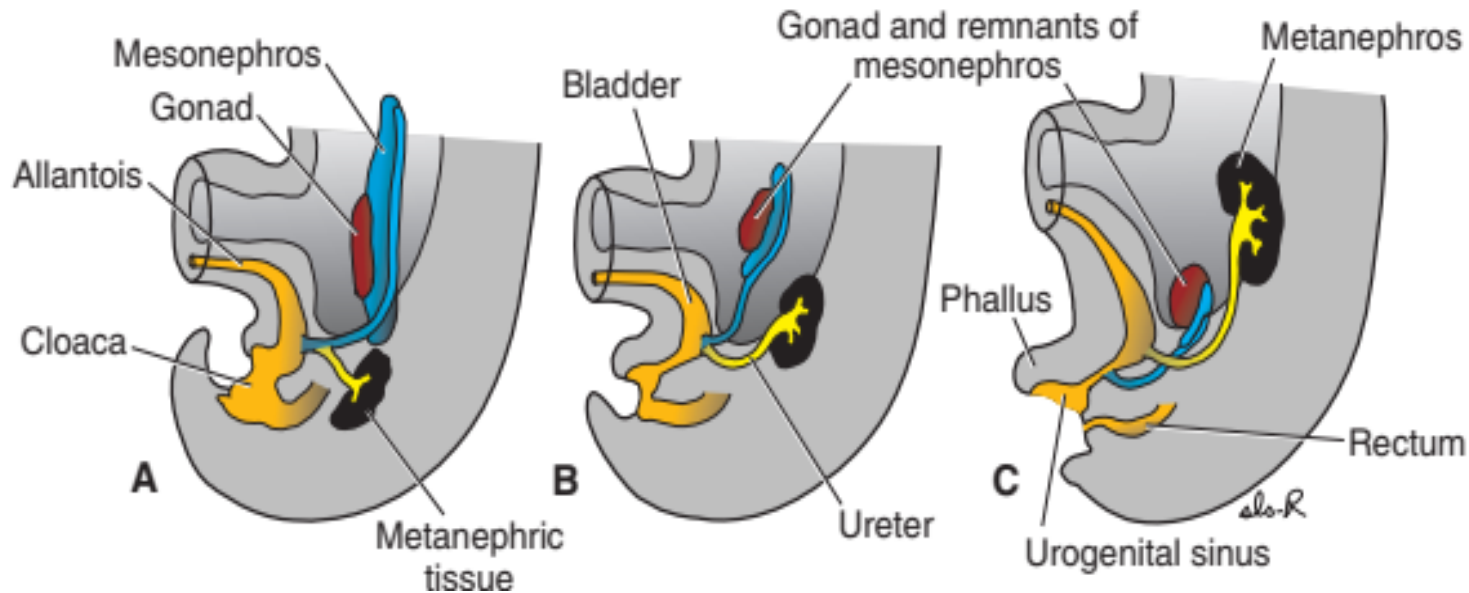


Figure 16.8. Surface view of a fetal kidney with multiple cysts characteristic of polycystic kidney disease.

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DEFINITIVE KIDNEY POSITION OF KIDNEY

- Initially, kidneys in the pelvis.
- With more development of lumbar and sacral regions, they shift cranially to abdomen.

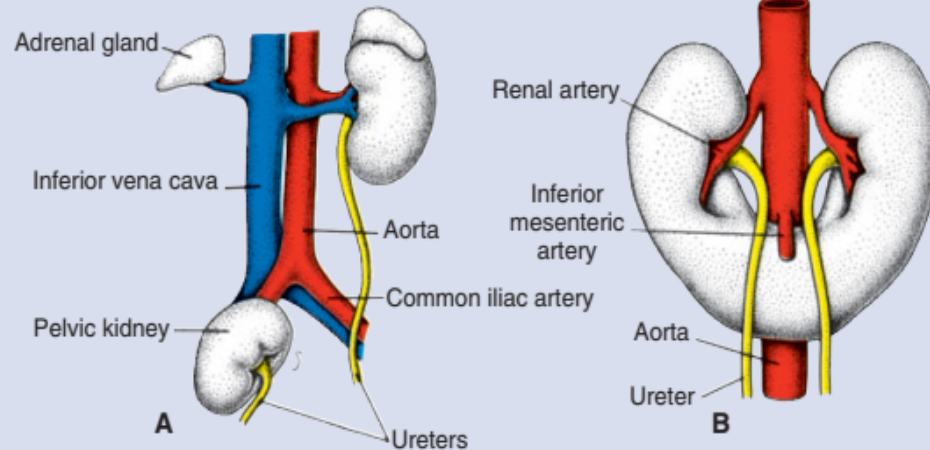


Abnormal Location of the Kidneys

During their ascent, the kidneys pass through the arterial fork formed by the umbilical arteries, but occasionally, one of them fails to do so. Remaining in the pelvis close to the common iliac artery, it is known as a **pelvic kidney** (Fig. 16.11A). Sometimes, the kidneys are pushed so close together during their passage through the arterial fork, that the lower poles fuse, forming a **horseshoe kidney** (Fig. 16.11B,C). The horseshoe kidney is usually at the level of the lower

lumbar vertebrae, since its ascent is prevented by the root of the inferior mesenteric artery (Fig. 16.11B). The ureters arise from the anterior surface of the kidney and pass ventral to the isthmus in a caudal direction. Horseshoe kidney is found in 1/600 people.

Accessory renal arteries are common; they derive from the persistence of embryonic vessels that formed during ascent of the kidneys. These arteries usually arise from the aorta and enter the superior or inferior poles of the kidneys.



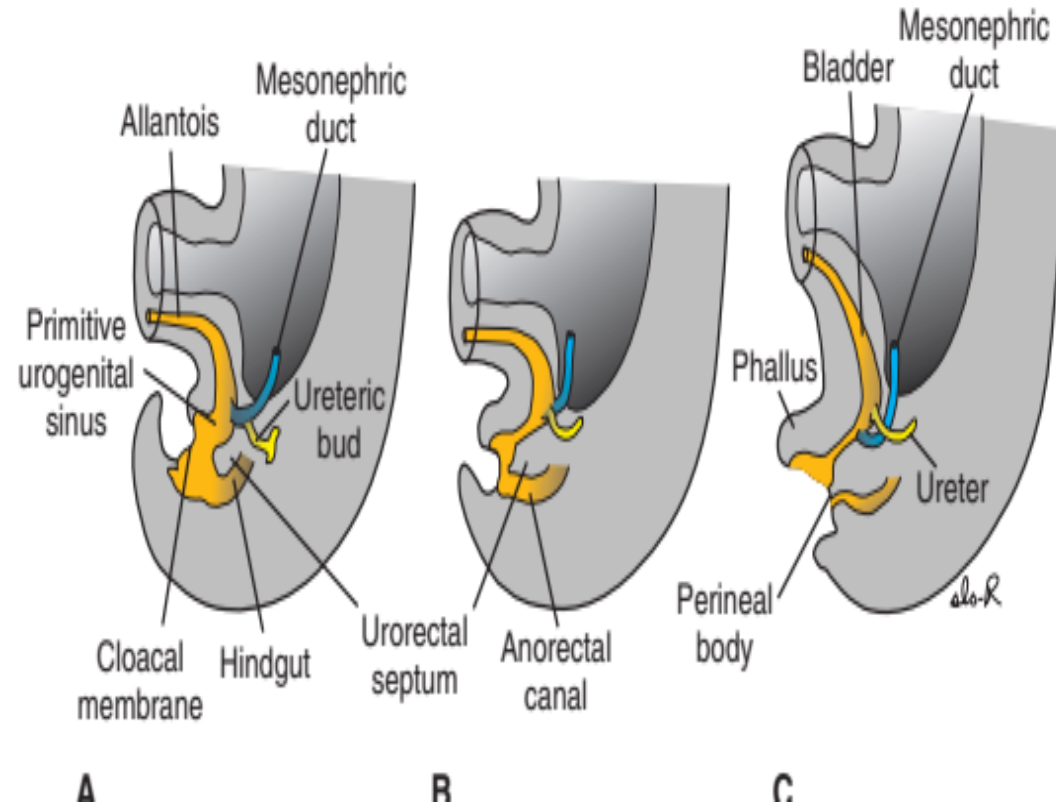
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Figure 16.11 A. Unilateral pelvic kidney showing the position of the adrenal gland on the affected side. B,C. Drawing and photomicrograph, respectively, of horseshoe kidneys showing the position of the inferior mesenteric artery.

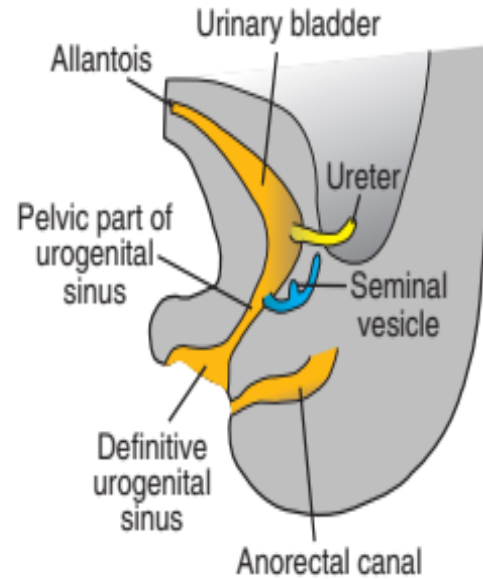
BLADDER & URETHRA

- During 4th to 7th week of development, cloaca divided into 2 parts:
 - ✓ Urogenital sinus: anteriorly.
 - ✓ Anal canal: posteriorly.
- Division occurs by:
 - ✓ Uro- rectal septum: a layer of mesoderm.
 - ✓ The tip of septum forms: perineal body.

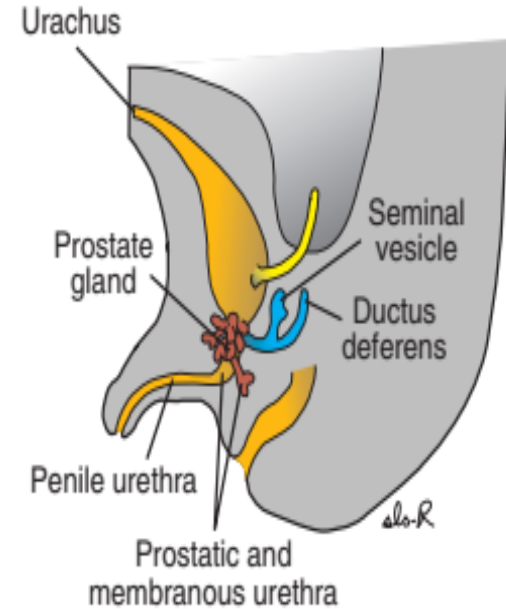


BLADDER & URETHRA

- Now, urogenital sinus divided into 3 parts:
 - a. Upper largest part: Urinary Bladder.
 - b. Middle narrow part: Pelvic part of UG sinus: in male gives rise to Prostatic & Membranous Urethra.
 - c. Lower (caudal) part: Phallic part: develops in genital system.



A

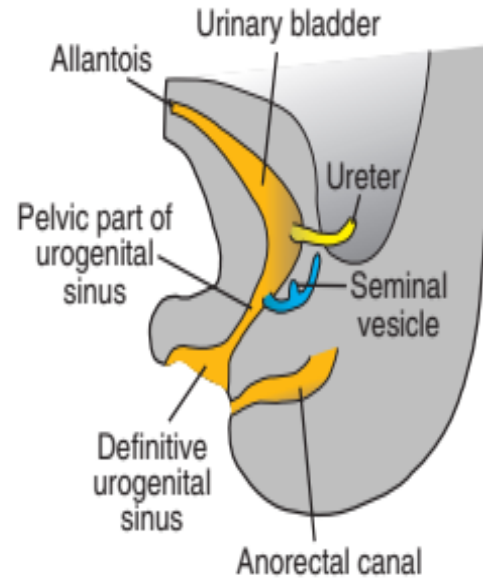


B

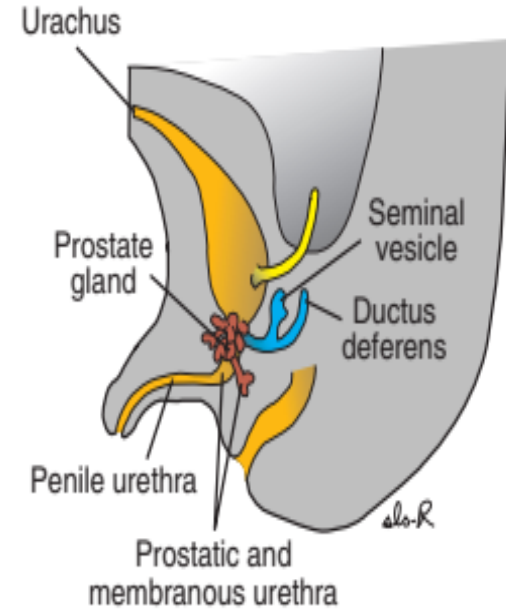


BLADDER & URETHRA

- Initially, bladder continuous with allantois.
- When allantois obliterated; it becomes thick fibrous cord (Urachus).
- Urachus connect bladder with umbilicus.
- It forms in adult: Median Umbilical Ligament.



A

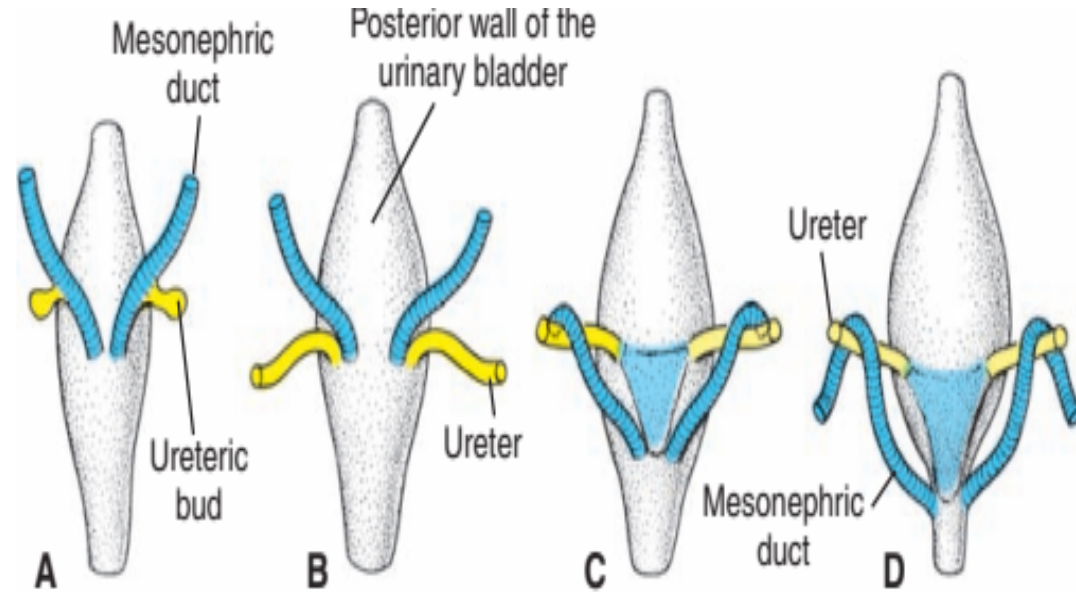


B



URETER

- During cloaca differentiation, caudal part of mesonephric ducts show small buds (Ureteric buds).
- U buds are absorbed into wall of bladder.
- During ascends of kidney, buds elongated to form ureters.
- Mesonephric ducts move close together & enter urethra (in male form ejaculatory ducts).



Bladder Defects

When the lumen of the intraembryonic portion of the allantois persists, a **urachal fistula** may cause urine to drain from the umbilicus (Fig. 16.15A). If only a local area of the allantois persists, secretory activity of its lining results in a cystic dilation, a **urachal cyst** (Fig. 16.15B). When the lumen in the upper part persists, it forms a **urachal sinus**. This sinus is usually continuous with the urinary bladder (Fig. 16.15C).

Exstrophy of the bladder (Fig. 16.16A) is a ventral body wall defect in which the bladder mucosa is exposed. Epispadias is a constant feature (Fig. 16.35), and the open urinary tract extends along the dorsal aspect of the penis through the bladder to the umbilicus. Exstrophy of the bladder is probably due to failure of the

lateral body wall folds to close in the midline in the pelvic region (see Chapter 7, p. 87). This anomaly is rare, occurring in 2/100,000 live births.

Exstrophy of the cloaca (Fig. 16.16B) is a more severe ventral body wall defect in which progression and closure of the lateral body wall folds are disrupted to a greater degree than is observed in bladder exstrophy (see Chapter 7, p. 87). In addition to the closure defect, normal development of the urorectal septum is altered, such that anal canal malformations and imperforate anus occur (see Chapter 15, p. 228). Furthermore, since the body folds do not fuse, the genital swellings are widely spaced resulting in defects in the external genitalia (Fig. 16.16B). Occurrence of the defect is rare (1/30,000).

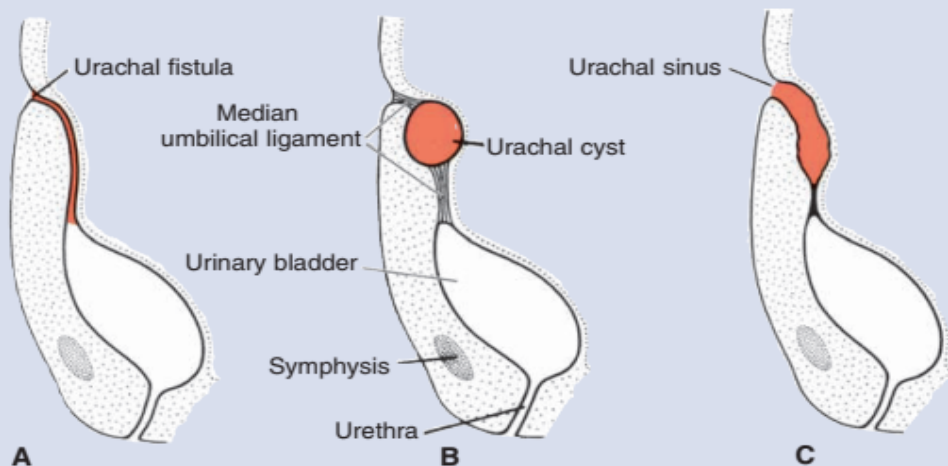


Figure 16.15 A. Urachal fistula. B. Urachal cyst. C. Urachal sinus. The sinus may or may not be in open communication with the urinary bladder.

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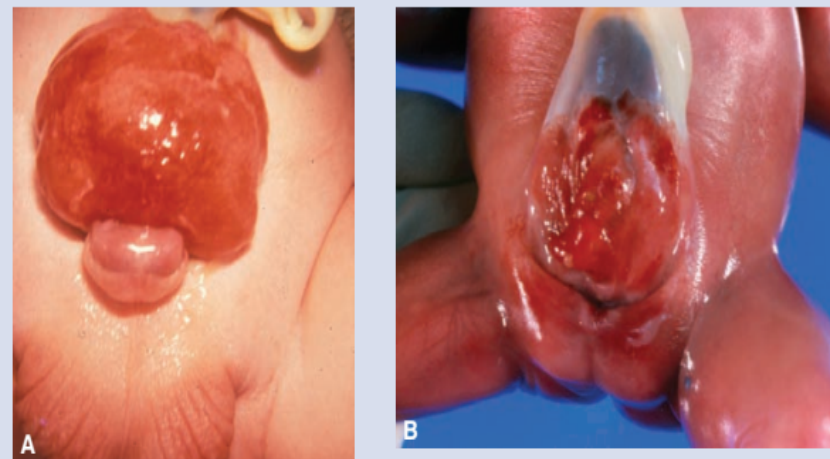


Figure 16.16 A. Exstrophy of the bladder. B. Cloacal exstrophy in a newborn.

THANKS

